Steering Diffusion Language Models via Policy Learning

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Abstract

Diffusion-based language models (DLMs) generate sequences through an iterative process of masked token prediction and remasking, where the choice of remasking schedule plays a critical role in determining sample quality. Existing approaches rely on fixed or heuristic strategies, such as low-confidence remasking or predefined schedules, which are not optimized for downstream objectives. In this work, we propose to replace these hand-designed schedules with a learned, adaptive policy that controls the diffusion trajectory at inference time. We formulate remasking as a sequential decision-making problem, where a policy observes per-token[1] features—including token identity, model confidence, entropy, and timestep, decides which tokens to keep or remask at each step. The policy is trained using reinforcement learning[2] to maximize task-specific rewards defined over global constraints, enabling structured generation without modifying the underlying diffusion language model. Our approach treats diffusion sampling as a controllable inference process, allowing a frozen backbone model to be steered toward improved outputs purely through policy optimization. Experiments on structured generation tasks demonstrate that learned remasking policies outperform fixed schedules and heuristic strategies. On 4×4 Sudoku using LLaDA-8B-Base, our approach improves the average solution score from 29.60 (confidence-based remasking) to 32.13 under 8 diffusion steps, corresponding to a +2.53 absolute gain. These improvements highlight that inference-time control via learned policies enables better constraint satisfaction and more efficient refinement. Our results establish a new direction for controllable generation in diffusion language models, where remasking decisions are learned rather than manually designed.

1 Introduction

Large language models (LLMs) are traditionally built upon the autoregressive paradigm, where tokens are generated sequentially in a left-to-right manner. Recently, diffusion-based approaches such as **LLaDA** [3] have challenged this formulation by introducing a masked diffusion language modeling framework, where generation is cast as an iterative denoising process over token sequences. In this setting, a forward process progressively masks tokens, while a learned mask predictor reconstructs them in parallel during the reverse process, enabling bidirectional reasoning and flexible generation dynamics.

A key but underexplored component of this framework is the *remasking schedule* used during inference. At each reverse diffusion step, a subset of predicted tokens may be remasked to maintain consistency with the forward process. Existing methods rely on fixed or heuristic strategies[4] (e.g., linear schedules or low-confidence remasking), treating the schedule as a hand-designed hyperparameter rather than an optimizable component of the generation process. However, this

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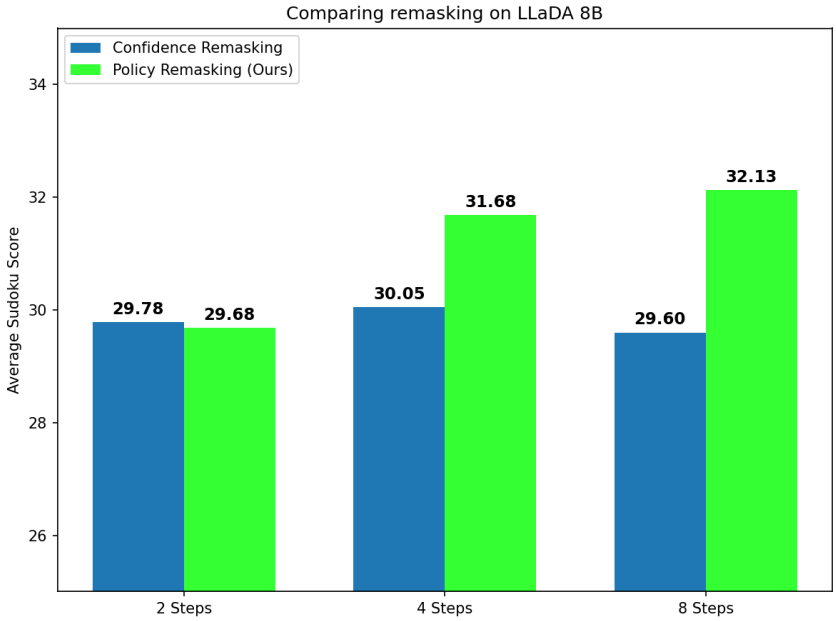

Figure 1: Average 4×4 Sudoku score (higher is better) using 2/4/8 diffusion steps. Policy-based remasking outperforms confidence remasking for LLaDA-8B-Base-Base in 2 of the 3 settings.

choice directly influences the quality of sampling, as it governs how uncertainty is resolved across diffusion steps.

In this work, we propose to replace this hand-crafted remasking schedule with a *learned, adaptive policy*[5]. Specifically, we formulate remasking as a sequential decision-making problem, where at each diffusion step, a policy determines which tokens to keep visible and which to remask. The objective is to guide the diffusion trajectory toward outputs that satisfy global, non-local constraints.

We study this problem in the context of 4×4 Sudoku, a structured generation task where validity depends on constraints spanning the entire sequence. Unlike standard language modeling benchmarks, Sudoku requires coordinated decisions across distant tokens, making it a suitable testbed for evaluating whether diffusion dynamics can be controlled to enforce global consistency. A learned policy can, in principle, discover strategies such as committing to high-confidence tokens early while deferring uncertain or conflicting positions to later steps, analogous to iterative refinement mechanisms in masked generation.

Beyond this specific task, our formulation raises a broader question: *can a frozen diffusion language model be steered toward improved reasoning and constraint satisfaction purely by controlling its sampling dynamics?* This perspective aligns with the probabilistic view of diffusion language models like LLaDA, where generation corresponds to iterative inference under a learned distribution, and suggests that performance gains may be achieved without modifying model parameters. Another point to consider is the time and compute needed to train a policy.

To this end, we introduce an RL-based remasking policy that operates on a compact per-token state, including token identity, model confidence (logits or probabilities), entropy, and timestep information. The policy is trained to maximize a task-specific reward that captures global constraint satisfaction, effectively replacing fixed remasking schedules with an adaptive, data-driven strategy.

Our contributions are the following:

- We formulate diffusion remasking as a reinforcement learning problem over the sampling trajectory of masked diffusion language models.
- We design a per-token policy that adaptively controls visibility during generation based on model uncertainty and temporal context.
- We empirically validate our approach on both LLaDA-1.5B and LLaDA-8B-Base, showing consistent improvements across model scales and demonstrating the effectiveness of learned

remasking policies. Specifically, using 8 diffusion steps during inference, our learned policy improves the average Sudoku score from **29.60** (LLaDA-8B-Base-Base with confidence remasking) to **32.13**, establishing a strong improvement without modifying the underlying model parameters. Please refer 1.

2 Related Work

Masked Diffusion Language Models. LLaDA introduces a diffusion-based alternative to autoregressive language modeling, where generation is framed as iterative masked token prediction. The motivation for this comes from attributing the success of autoregressive models to the transformer architecture rather than the sequential token generation. Specifically, a forward process masks tokens independently with a ratio $t \sim \mathcal{U}(0, 1)$, while a Transformer-based mask predictor models the reverse process by reconstructing masked tokens without a causal constraint. At inference, generation proceeds from a fully masked sequence ($t = 1$) to an unmasked output ($t = 0$) through a discretized reverse trajectory. A key component of this process is *remasking*, where a subset of predicted tokens is masked again at each step to maintain consistency with the forward process. While random remasking is theoretically grounded, LLaDA adopts a low-confidence remasking heuristic, inspired by annealing strategies, highlighting the importance of schedule design in diffusion sampling.

Iterative Masked Generation. A closely related paradigm is MaskGIT [6], which performs parallel token prediction with iterative refinement. At each iteration, MaskGIT retains high-confidence predictions and remasks the remaining tokens, following a predefined schedule (e.g., cosine). This demonstrates that generation quality is highly sensitive to the choice of refinement schedule. Our work can be viewed as a generalization of this idea: instead of relying on fixed heuristics, we learn a token-wise remasking policy that adaptively determines which tokens to keep or revise.

Schedule Optimization in Diffusion. Recent work has highlighted that the sampling schedule itself is a critical factor in diffusion performance. For instance, methods such as *Jump Your Steps* [7] optimize timestep allocation to mitigate error accumulation under limited sampling budgets. These results support the broader perspective that schedules should be treated as optimization targets rather than fixed design choices, motivating our formulation of remasking as a learnable policy.

Reinforcement Learning for Control. We formulate remasking as a sequential decision-making problem and optimize it using policy gradient methods. REINFORCE [8] provides an unbiased estimator for policy gradients, typically combined with baselines for variance reduction. For improved stability, extensions such as Proximal Policy Optimization (PPO) [9] can be employed, which optimize a clipped surrogate objective over multiple updates. In our setting, RL enables learning an adaptive controller over diffusion dynamics, without modifying the underlying language model.

RL Fine-Tuning for Diffusion Language Models. Recently, [10] introduced $d1$, a framework that adapts pre-trained masked diffusion language models into reasoning models through a combination of supervised fine-tuning and a novel policy-gradient RL algorithm called diffu-GRPO. Notably, they evaluate their method on structured reasoning and planning tasks, including Countdown and Sudoku, demonstrating that RL can significantly enhance the multi-step reasoning capabilities of diffusion models. While diffu-GRPO focuses on updating the model parameters directly via policy gradients, our work takes a complementary approach: we keep the language model backbone frozen and instead learn a dedicated remasking policy to steer the inference-time generation trajectory.

3 Masked Diffusion Language Models

Unlike autoregressive models, which factorize the data distribution as a product of conditional probabilities, LLaDA defines a generative model $p_\theta(x_0)$ through a forward masking process and a learned reverse denoising process.

Forward Process. Given a sequence $x_0 \in \mathcal{V}^L$, the forward process[11] produces a corrupted sequence x_t by independently masking each token with probability $t \in [0, 1]$. Formally, for each

position i:

$$x_{t,i} = \begin{cases} \texttt{[MASK]} & \text{with probability } t, \\ x_{0,i} & \text{with probability } 1-t. \end{cases} \tag{1}$$

Thus, x_t contains a random subset of masked tokens, and at $t = 1$, all tokens are masked.

Reverse Process. The reverse process is parameterized by a mask predictor $p_\theta(\cdot \mid x_t)$, which predicts the original tokens at all masked positions simultaneously. Let $\mathcal{M}_t = \{i \mid x_{t,i} = \texttt{[MASK]}\}$ denote the masked positions. The model defines:

$$p_\theta(x_0 \mid x_t) = \prod_{i \in \mathcal{M}_t} p_\theta(x_{0,i} \mid x_t), \tag{2}$$

where each conditional distribution is given by a softmax over vocabulary logits.

Training Objective. The model is trained using a masked cross-entropy loss over randomly sampled corruption levels:

$$\mathcal{L}(\theta) = -\mathbb{E}_{x_0,\, t,\, x_t} \left[\frac{1}{t} \sum_{i=1}^{L} \mathbf{1}[x_{t,i} = \texttt{[MASK]}] \log p_\theta(x_{0,i} \mid x_t) \right], \tag{3}$$

where $t \sim \mathcal{U}(0,1)$ and $x_t \sim q(x_t \mid x_0)$. The $\frac{1}{t}$ reweighting corrects for the expected number of masked tokens and ensures proper scaling across corruption levels.

This objective has been shown to upper bound the negative log-likelihood of the model distribution:

$$-\mathbb{E}_{p_{\text{data}}(x_0)} \left[\log p_\theta(x_0) \right] \leq \mathcal{L}(\theta), \tag{4}$$

providing a principled maximum likelihood training objective.

Iterative Generation. At inference, generation proceeds by initializing from a fully masked sequence x_1 and iteratively refining it by decreasing t toward 0. At each step, the model predicts token distributions for masked positions and updates the sequence accordingly.

A critical component of this process is the *remasking strategy*. After each prediction step, a subset of tokens is re-masked before proceeding to the next iteration. Existing approaches rely on heuristic rules, typically based on model confidence. For example, defining token confidence as

$$c_{t,i} = \max_{v \in \mathcal{V}} p_\theta(v \mid x_t), \tag{5}$$

tokens with lowest confidence are selected for remasking.

Limitations. Such confidence-based heuristics are fixed and operate independently per token, ignoring global structure and task-specific constraints. As a result, they may fail to guide the generation process effectively in structured settings where token dependencies are critical.

Towards Learned Remasking. This motivates our approach: instead of relying on hand-designed confidence-based rules, we propose to learn a policy that adaptively determines which tokens to remask at each step, enabling task-aware and globally consistent refinement of the diffusion trajectory.

4 Proposed Remasking Policy Network

We replace heuristic remasking schedules in masked diffusion language models with a learned policy trained via reinforcement learning. The diffusion backbone (e.g., LLaDA) is kept frozen, and a lightweight policy controls the sampling trajectory by deciding which tokens to keep or remask at each step.

State Representation. At diffusion step t, for a sequence of N tokens, we construct a per-token state:

$$S_t = \{s_{t,i}\}_{i=1}^{N}, \quad s_{t,i} = \big[e(x_{t,i});\ p_{t,i};\ H_{t,i};\ \alpha_t\big],$$

where $e(x_{t,i})$ is the embedding of the current token (including mask state), $p_{t,i} \in \mathbb{R}^K$ are normalized probabilities over valid tokens (e.g., $K = 4$ for Sudoku), $H_{t,i} = -\sum_v p(v) \log p(v)$ is the entropy, and $\alpha_t = t/T$ is the normalized timestep.

Policy Network. A lightweight encoder f_θ (e.g., Transformer) produces contextual representations:

$$H_t = f_\theta(S_t) \in \mathbb{R}^{N \times d}.$$

The policy outputs per-token keep probabilities:

$$\pi_\theta(a_{t,i} = 1 \mid S_t) = \sigma(w^\top h_{t,i} + b),$$

where $a_{t,i} \in \{0, 1\}$ indicates whether token i is kept (unmasked). Actions are sampled independently with constraints enforced for fixed tokens.

Controlled Sampling. Given actions a_t, tokens with $a_{t,i} = 0$ are remasked before the next diffusion step. This replaces fixed schedules with adaptive control over the reverse diffusion process.

Task-Specific Reward (Sudoku). Let $\hat{x}^{(t)} \in \{1, \ldots, L\}^{L \times L}$ denote the decoded board at step t. We define a constraint violation score:

$$E(\hat{x}) = \sum_{r=1}^{L}(L - |\mathrm{Unique}(\hat{x}_{r,:})|) + \sum_{c=1}^{L}(L - |\mathrm{Unique}(\hat{x}_{:,c})|) + \sum_{g=1}^{L}(L - |\mathrm{Unique}(\hat{x}_g)|)\,, \quad (6)$$

where $\mathrm{Unique}(\cdot)$ counts distinct values and $\hat{x}_g$ indexes subgrids. A valid solution satisfies $E(\hat{x}) = 0$.

We define a dense improvement reward:

$$r_t = E(\hat{x}^{(t-1)}) - E(\hat{x}^{(t)}), \quad (7)$$

which is positive when constraint violations decrease and negative otherwise.

In our experiments, we instantiate this framework on $L = 4$ Sudoku.

Return and Optimization. The discounted return at step t is defined as $R_t = \sum_{k=0}^{T-t} \gamma^k r_{t+k}$. The policy is optimized using REINFORCE with a baseline. The reason we chose this algorithm is because we wanted a lightweight policy to use at inference time. REINFORCE, with no value network needed for policy evaluation, fits this setting well. $\mathcal{L}(\theta) = -\sum_{t=1}^{T} \log \pi_\theta(a_t \mid S_t)\,(R_t - b)$, where b is a variance-reduction baseline. The diffusion model remains frozen, and learning occurs solely in the policy.

5 Experimental Settings

Data. We evaluate our approach on the 4×4 Sudoku puzzle benchmark, a structured generation task with strict global constraints. Each instance consists of a partially filled grid $\hat{x} \in \{1, \ldots, 4\}^{4 \times 4}$ with a subset of entries provided as fixed clues and the remaining positions masked. The objective is to generate a valid completion satisfying row, column, and subgrid constraints. We use a synthetic dataset of randomly generated Sudoku puzzles with varying difficulty levels, ensuring diversity in clue patterns and solution structures. This controlled setting provides a well-defined testbed for studying constraint-aware generation and policy-guided refinement. In this setting, only four tokens out of the base model's vocabulary have to be passed along for policy training, significantly reducing the number of parameters needed for the policy network. This also allows for a smaller training dataset(100 samples taken from [12]) and faster training times, allowing for training in compute-constrained environments.

Models. We conduct experiments using two pretrained diffusion language models: LLaDA-1.5B and LLaDA-8B-Base. LLaDA is a masked diffusion language model that generates sequences via iterative denoising, where tokens are repeatedly predicted and selectively remasked over multiple steps. The 1.5B and 8B variants differ in scale and representational capacity, allowing us to evaluate the robustness of our approach across model sizes. In all experiments, the diffusion models are kept frozen, and only the lightweight remasking policy is trained, enabling us to isolate the effect of adaptive inference control.

Model	Remasking Strategy	Diffusion Steps	Accuracy (%)
LLaDA-1.5B	Confidence-Based	2	27.05
		4	27.80
		8	27.45
	Policy-Based (16 steps)	2	28.78
		4	**29.65**
		8	29.43
LLaDA-8B-Base	Confidence-Based	2	29.78
		4	30.05
		8	29.60
	Policy-Based (16 steps)	2	29.68
		4	31.68
		8	**32.13**

Table 1: Main results comparing confidence-based remasking with our learned policy-based remasking. All policy networks are trained for 200 epochs. LLaDA-1.5B and LLaDA-8B-Base results demonstrate consistent improvements across inference steps, with the largest gain observed at 4 and 8 diffusion steps respectively.

6 Results

6.1 Main Results

We evaluate our proposed policy-guided remasking framework on 4×4 Sudoku using pretrained LLaDA models of different scales. The policy network is trained to adaptively control token visibility during reverse diffusion, replacing fixed confidence-based remasking heuristics. Importantly, the diffusion backbone remains frozen, and all improvements stem solely from learned inference-time control. Results are summarized in Table 1. For LLaDA-1.5B and LLaDA-8B-Base, our method consistently outperforms the confidence-based baseline across all inference horizons. Notably, at 8 diffusion steps for LLaDA-8B-Base, accuracy improves from 29.60% to **32.13%**, demonstrating that a lightweight policy can significantly enhance structured generation without modifying model parameters. Similar gains are observed at intermediate steps, indicating that the learned policy effectively balances exploration and refinement during the diffusion process.

These findings highlight that adaptive remasking provides a powerful mechanism for steering diffusion-based language models toward better solutions under global constraints.

In the following section, we further analyze our method through ablations on policy training configurations, including varying the number of reverse diffusion steps and training duration, to better understand the factors driving these improvements.

6.2 Ablations

Different Steps for Policy Learning. We study the effect of training the remasking policy with varying diffusion horizons. Specifically, we train policy networks on 4×4 Sudoku using LLaDA-1.5B with training diffusion steps $T \in \{4, 8, 16\}$. At evaluation time, we test each learned policy under shorter inference horizons with $T \in \{2, 4, 8\}$ to assess generalization across sampling lengths. Results are reported in Table 2. We observe that policies trained with fewer steps (e.g., $T = 4$) perform poorly across all inference settings, indicating insufficient capacity to learn meaningful refinement strategies. Increasing the training horizon to $T = 8$ improves performance, but remains below the confidence-based baseline. In contrast, the policy trained with $T = 16$ significantly outperforms all other variants and even surpasses the confidence-based remasking baseline across all inference settings. Notably, it achieves the best performance at $T = 4$ inference (29.05%), demonstrating strong generalization despite mismatch between training and inference horizons. These results suggest that longer training trajectories enable the policy to learn more robust and transferable refinement strategies, which can be effectively applied even under constrained inference budgets.

Remasking Strategy	Diffusion Steps	Accuracy (%)
Confidence-Based	2	27.05
	4	27.80
	8	27.45
Policy-Based (4 steps)	2	7.70
	4	9.28
	8	10.83
Policy-Based (8 steps)	2	15.45
	4	17.80
	8	18.73
Policy-Based (16 steps)	**2**	**28.33**
	4	**29.05**
	8	**28.90**

Table 2: Sudoku accuracy across different remasking strategies for LLaDA 1.5B.

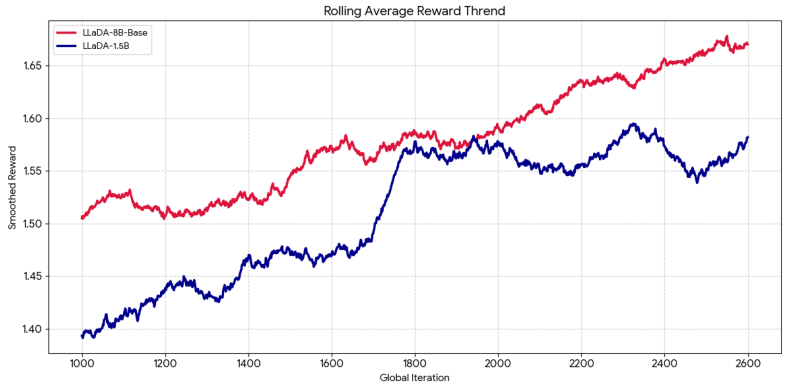

Figure 2: Rolling average reward during training for LLaDA-1.5B and LLaDA-8B-Base. Both models exhibit consistent improvement as training progresses, indicating stable policy learning.

Effect of Training Duration. We analyze the impact of policy training duration by comparing models trained for 20 and 200 epochs, while keeping the training diffusion horizon fixed at $T = 4$. Results are shown in Table 3. We observe consistent improvements across all inference settings as training progresses. For example, at 8 inference steps, accuracy improves from 10.83% (20 epochs) to 17.85% (200 epochs), indicating that longer training allows the policy to better capture refinement dynamics. This trend is further supported by the reward trajectories shown in Figure 2, where both LLaDA-1.5B and LLaDA-8B-Base-Base exhibit steady increases in average reward as training proceeds. Notably, the improvement is consistent across model scales, suggesting that policy learning continues to refine token selection strategies over time.

Overall, these results demonstrate that policy optimization remains effective beyond early training and that extended training leads to more robust and higher-quality generation.

7 Discussion & Future Works

Our results demonstrate that remasking plays a critical role in the performance of diffusion-based language models, and that replacing heuristic schedules with a learned policy provides a simple yet effective mechanism for improving structured generation. By operating entirely at inference time and keeping the underlying diffusion model frozen, our approach offers a lightweight and modular way to enhance existing models without retraining or architectural changes. A key insight from our experiments is that longer policy training horizons lead to more robust and transferable remasking strategies. Policies trained with larger diffusion steps are able to generalize effectively to shorter inference horizons, suggesting that they learn a richer notion of iterative refinement. Additionally,

Epochs	Diffusion Steps	Accuracy (%)
20	2	7.70
20	4	9.28
20	8	10.83
200	2	10.28
200	4	15.13
200	8	17.85

Table 3: Effect of training duration on policy performance. The policy is trained with 4 diffusion steps and evaluated under inference horizons of 2, 4, and 8 steps.

the observed improvements across both LLaDA-1.5B and LLaDA-8B-Base-Base indicate that this approach scales with model capacity and is not tied to a specific model size.

However, several limitations remain. First, our evaluation is restricted to a synthetic 4×4 Sudoku setting, which, while useful for controlled analysis, does not fully capture the complexity of real-world language generation tasks. Second, the current reward function relies on explicit, task-specific constraint verification, which may not be readily available in more open-ended domains. Third, policy training introduces additional computational overhead, particularly for longer diffusion horizons. There are several promising directions for future work. Extending this framework to larger and more complex structured tasks (e.g., 9×9 Sudoku or combinatorial reasoning benchmarks) would provide further evidence of its generality. Incorporating learned or proxy reward functions could enable application to tasks without explicit constraint checkers. Additionally, exploring more advanced reinforcement learning algorithms, such as PPO or GRPO, may improve training stability and sample efficiency.

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